Hands-on Lab: Network Protocol Analyzers

Estimated time needed: **45** minutes

About This Lab

In this lab, we will be using Wireshark. Wireshark is an open-source app that can capture and display data that reads the contents of each packet as it travels across the network.

Objectives

In this hands-on lab, you will:

- Install Wireshark.
- Capture packets using Wireshark.
- Review capture results.

Important Notices about This Lab

About Lab Sessions

Lab sessions are not persisted. This means that every time you connect to this lab, a new environment is created for you. Any data or files you saved in a previous session are no longer available. To avoid losing your data, plan to complete these tasks in a single session.

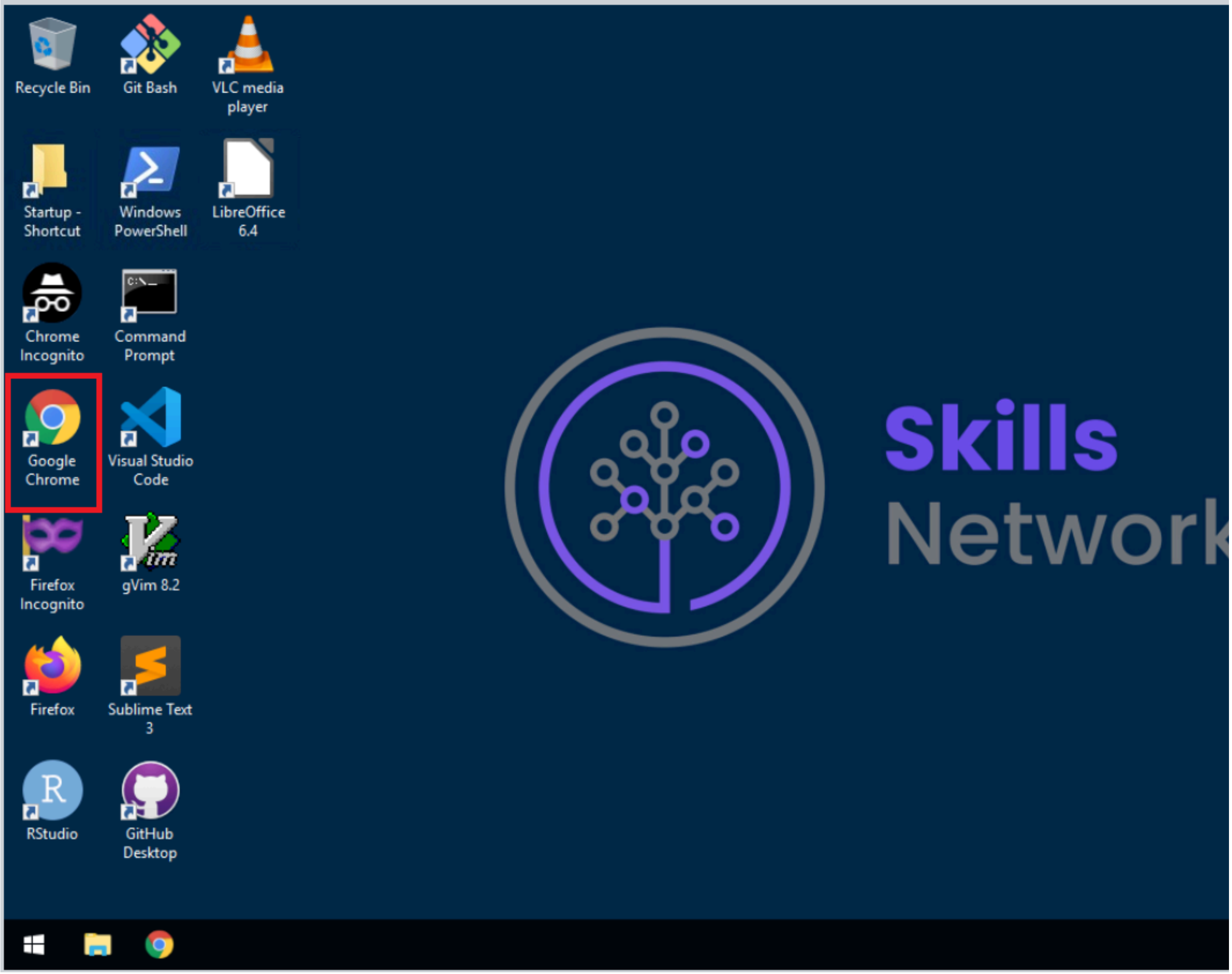
About the Lab Instructions and Solutions

Microsoft Windows operating system features can vary based on the Windows edition. If completing these exercises on your machine, your navigation and solutions may differ from what's presented in this lab.

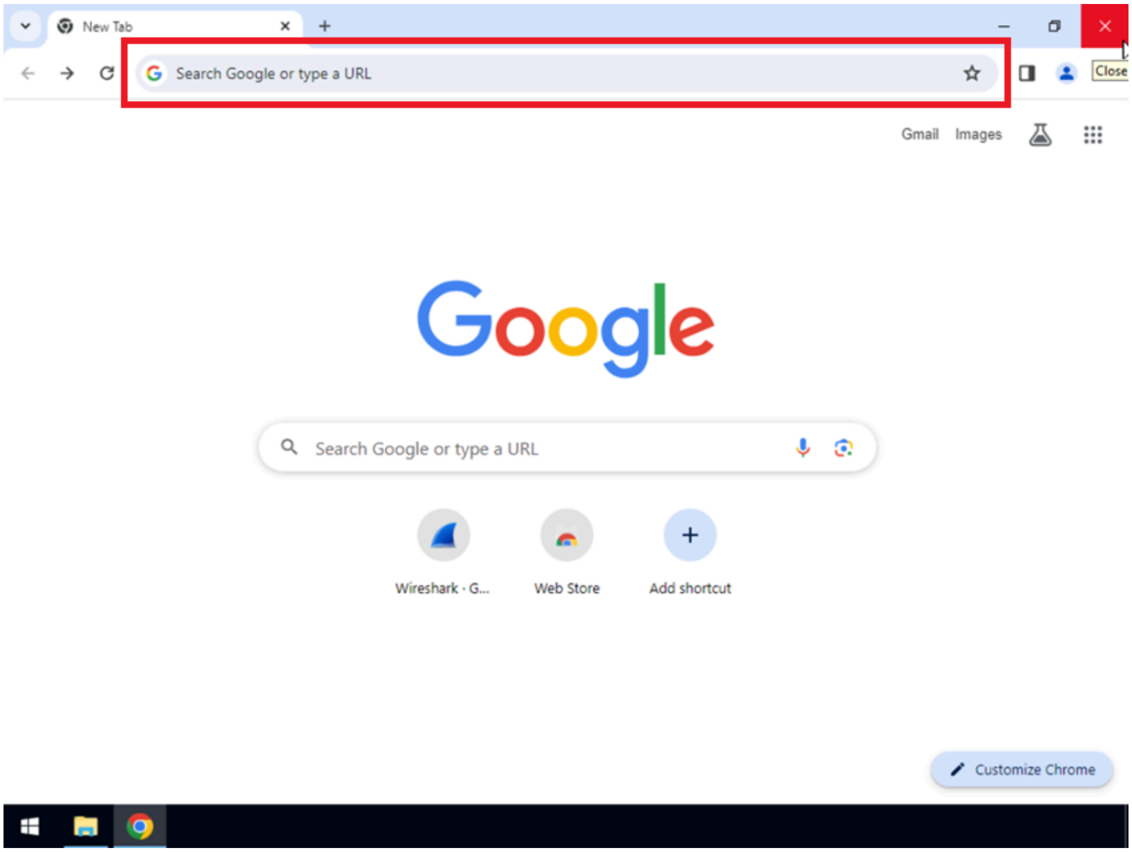
Exercise 1: Install Wireshark

In this exercise we will install Wireshark.

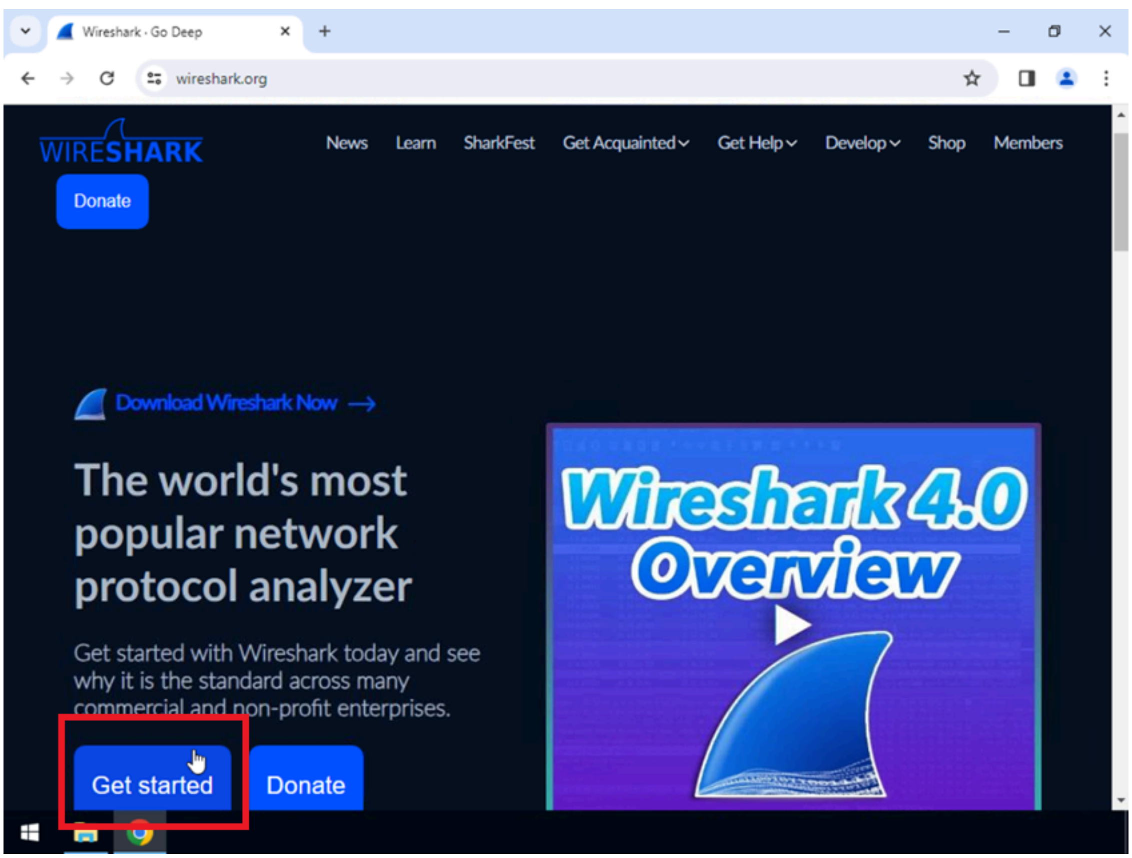
1. Open Chrome browser.



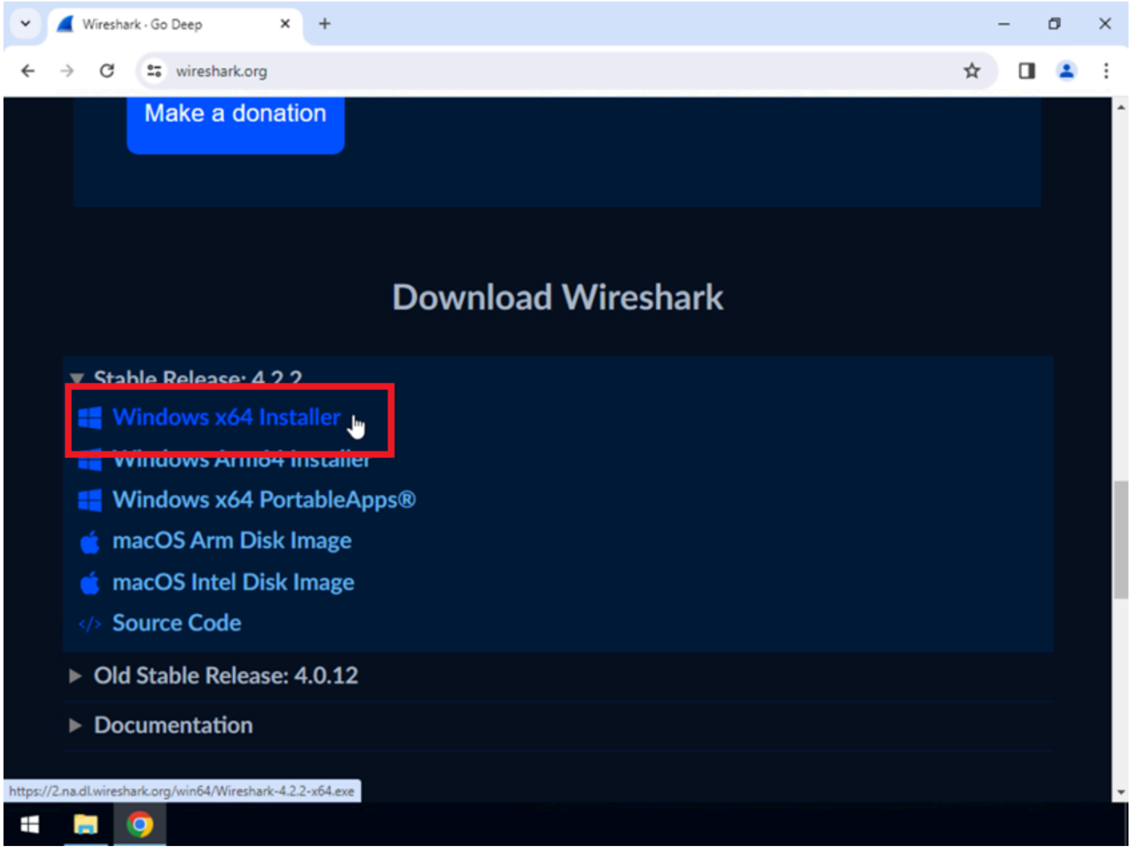
2. Navigate to **wireshark.org**.



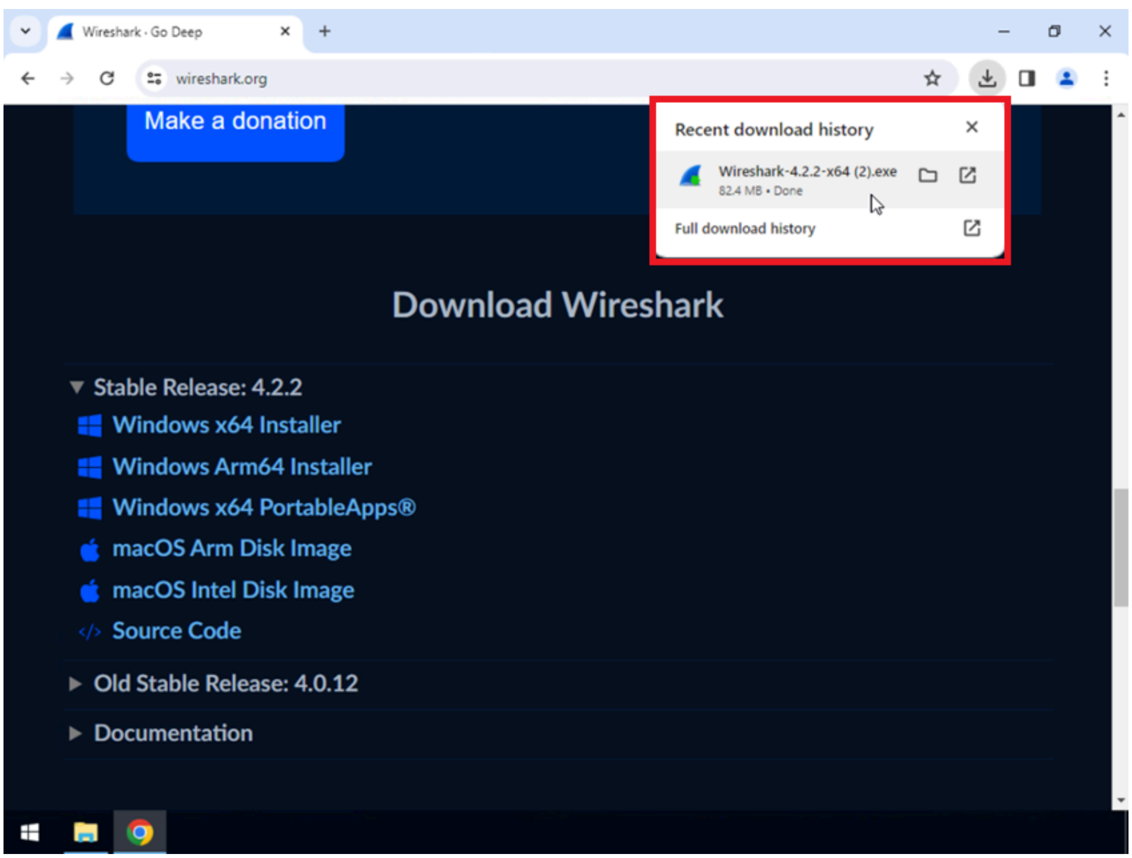
3. Click **Get started**.



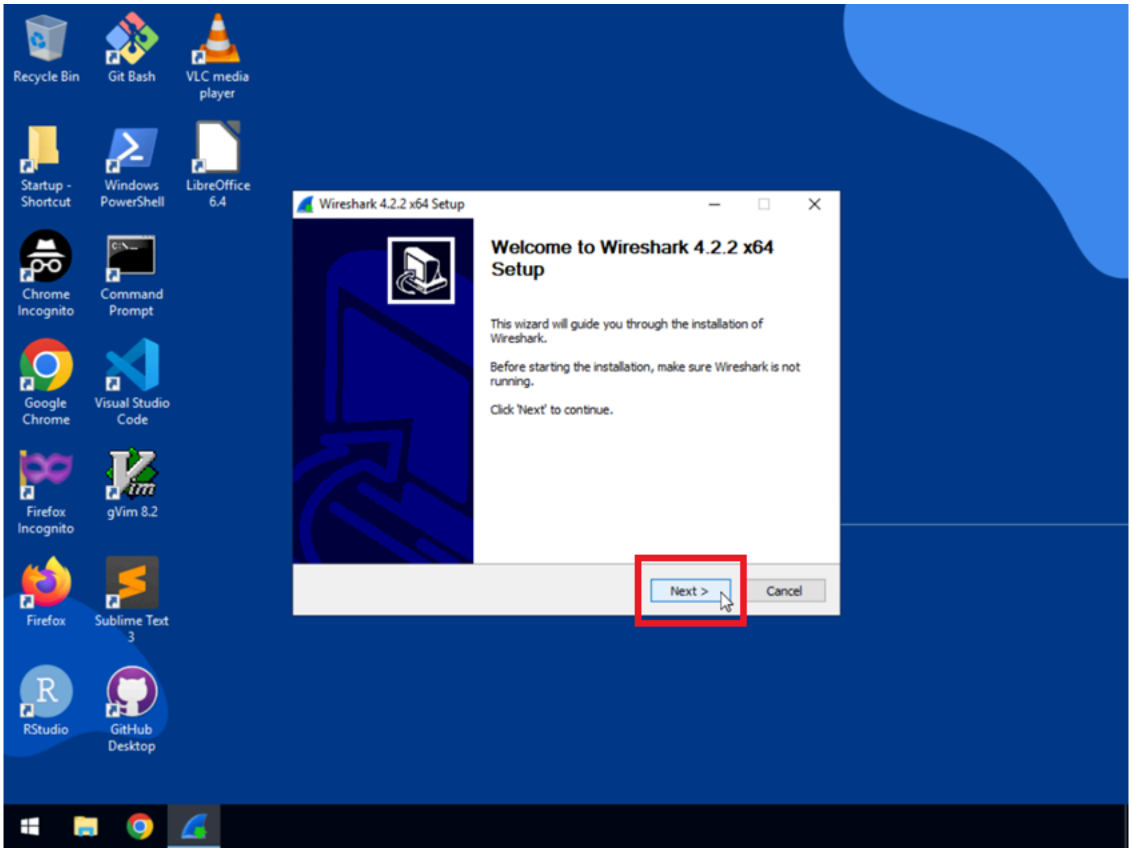
4. Click **Windows x64 Installer**.



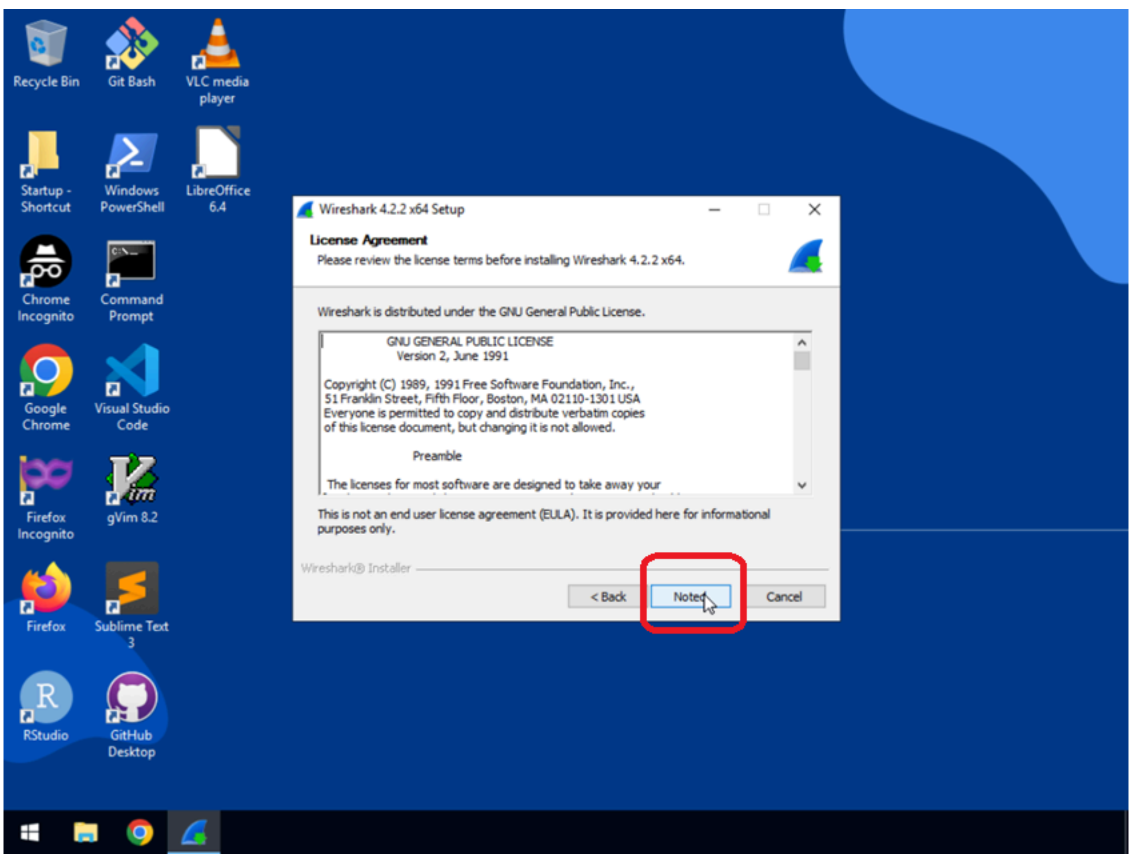
5. CLick the wireshark executable file to start the installation wizard.



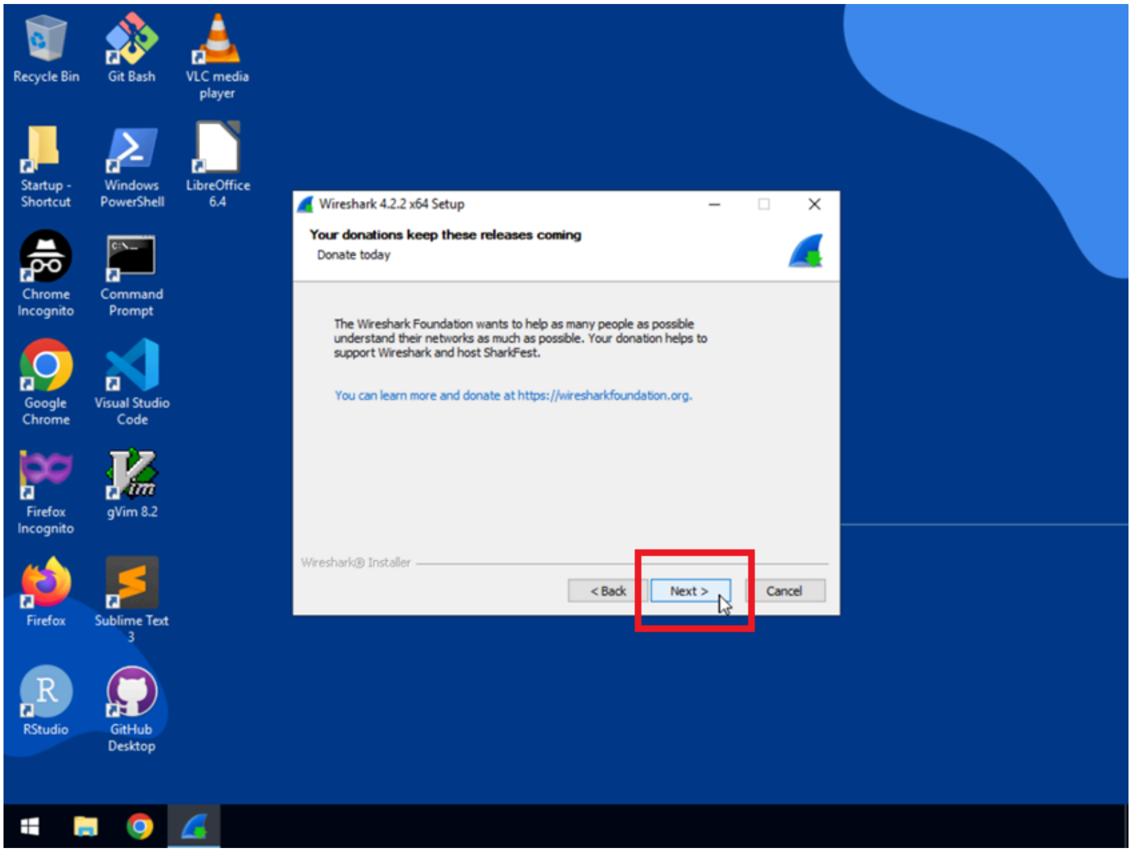
6. Click **Next** to continue.



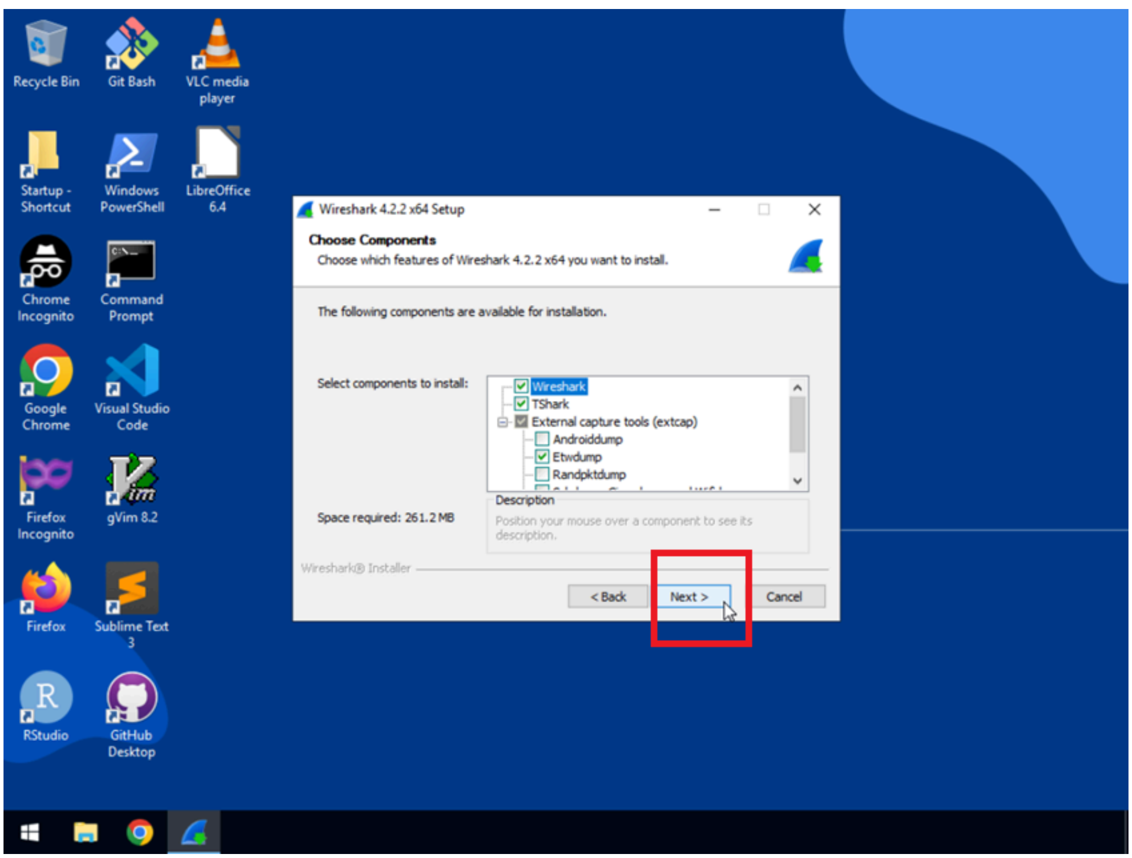
7. Click **Noted** to acknowledge licensing guidelines.



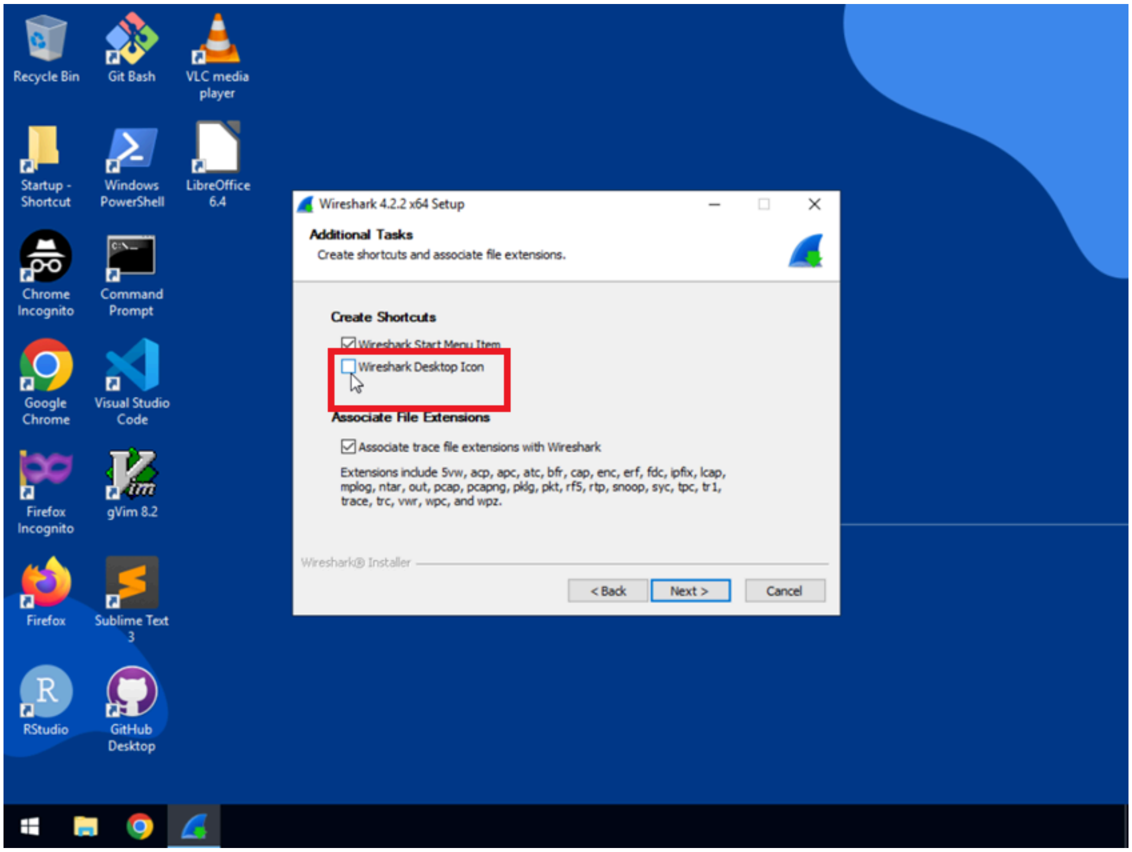
8. Click **Next** to continue.



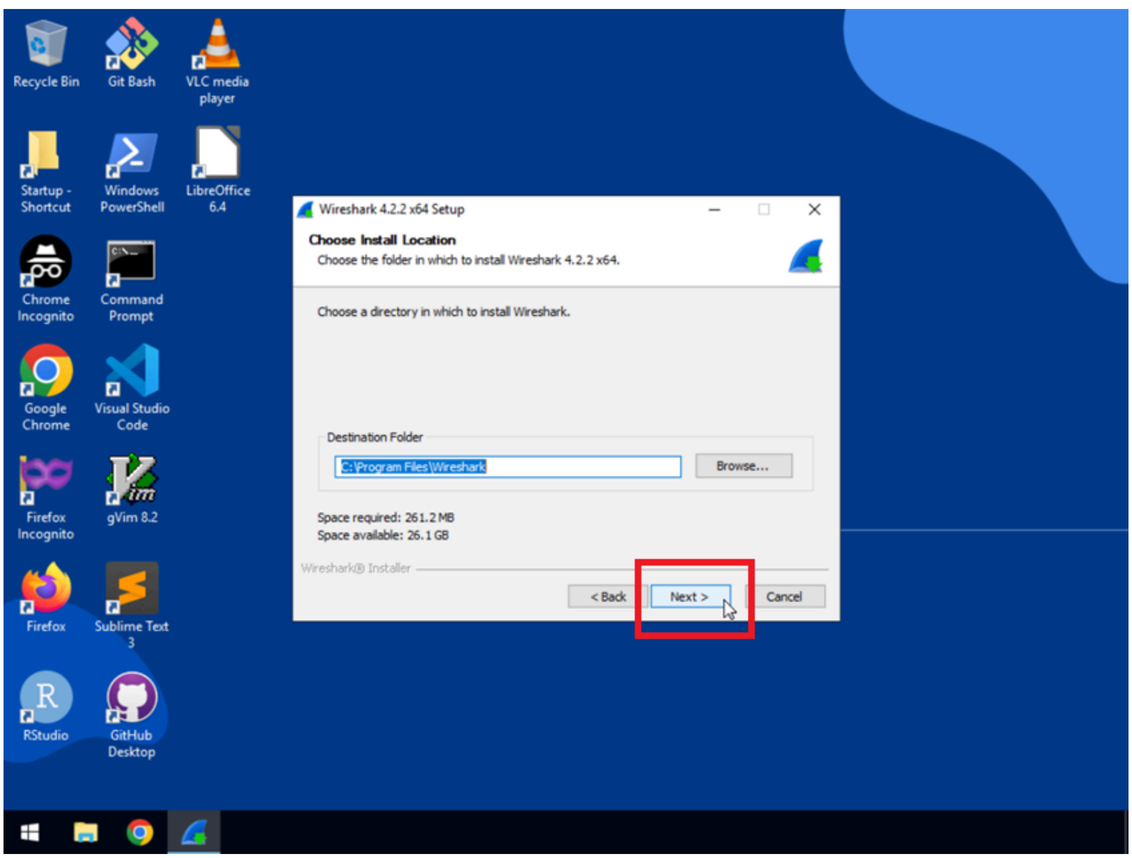
9. Click **Next** to keep the recommended components to install.



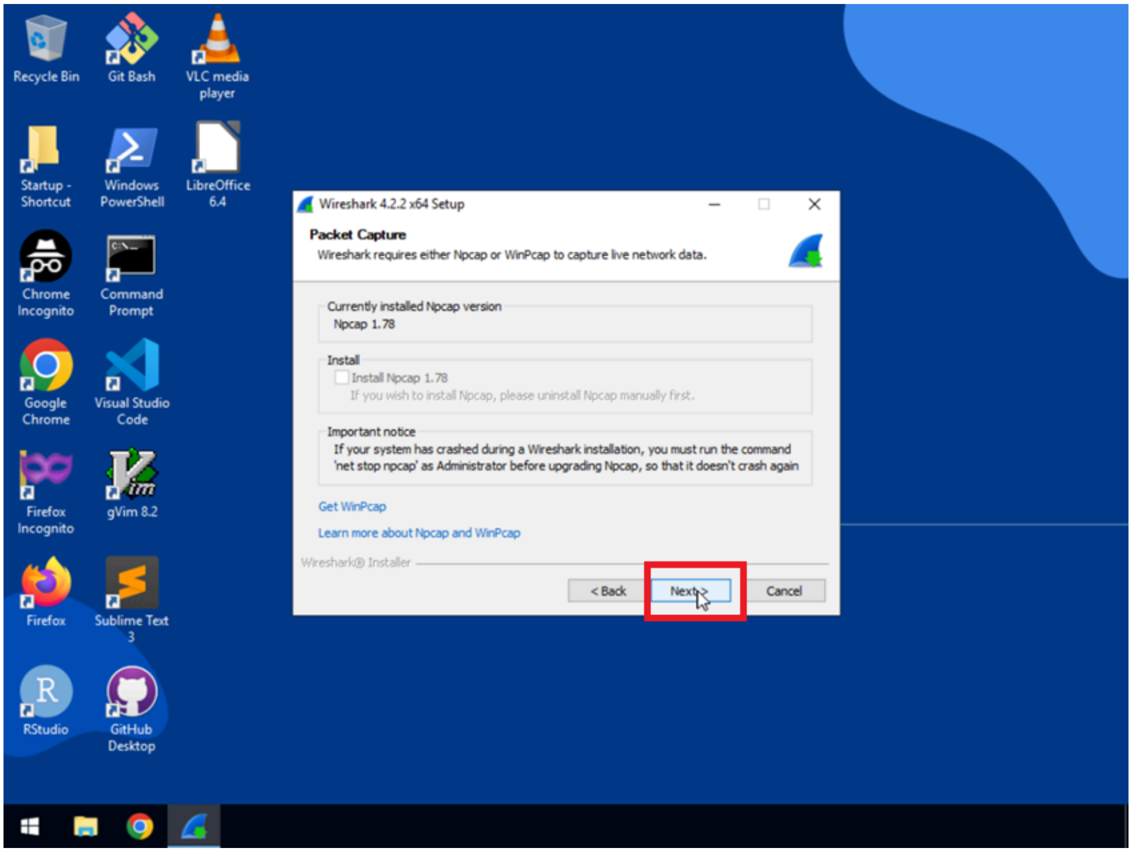
10. Click the box next to **Wireshark Desktop Icon**, then click **Next** to continue.



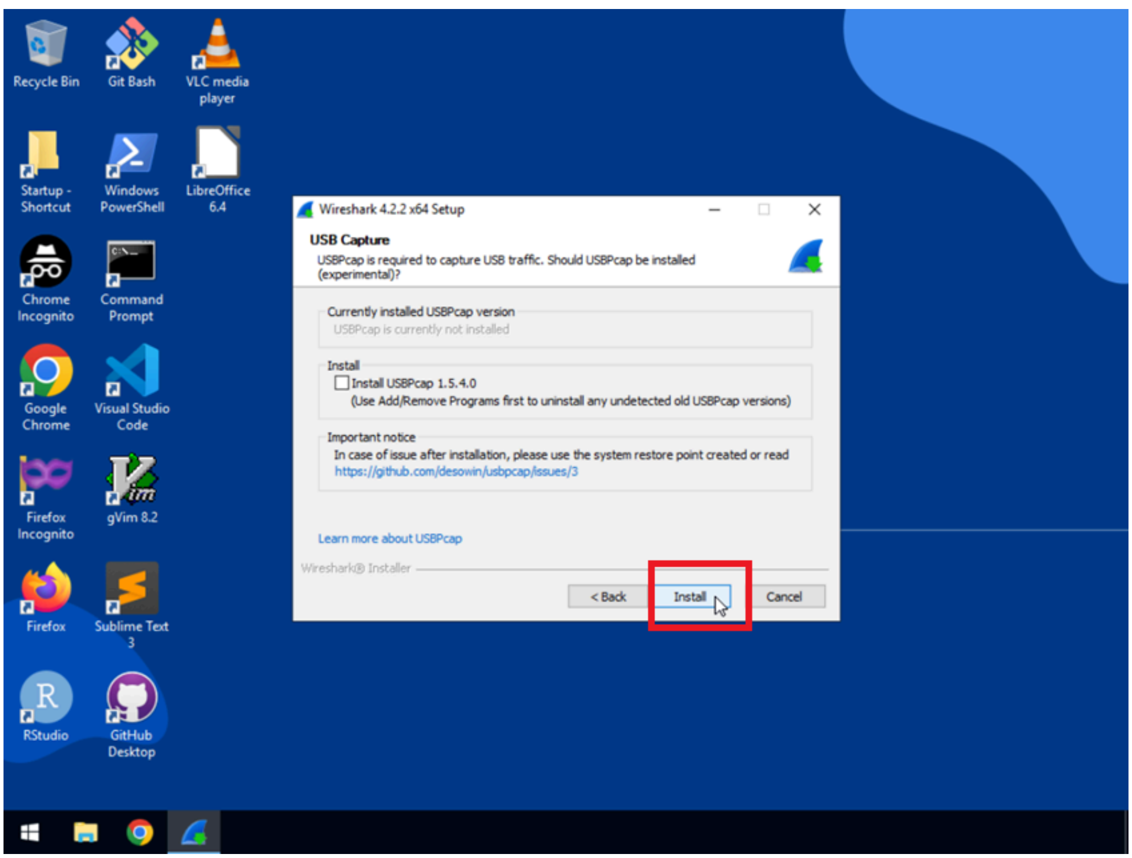
11. Click **Next** to keep the default destination folder as C:\Program Files\Wireshark.



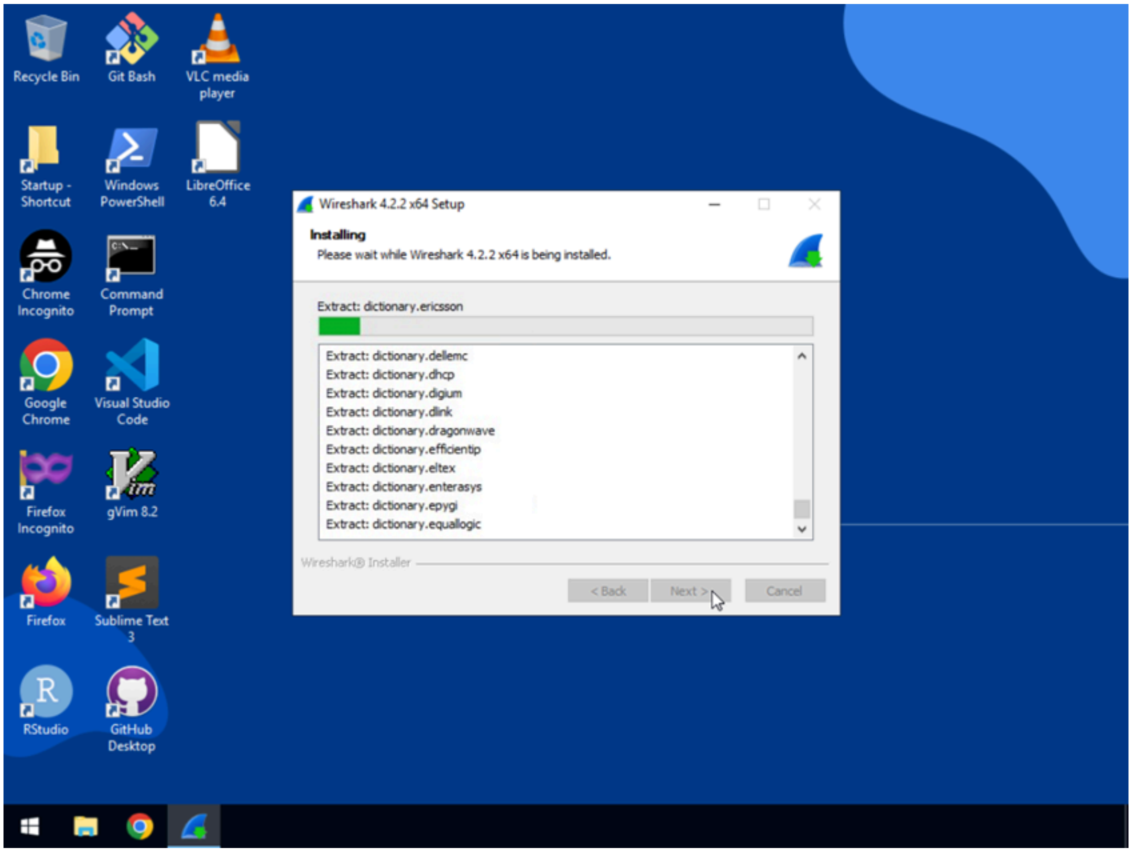
12. Click **Next** to continue.



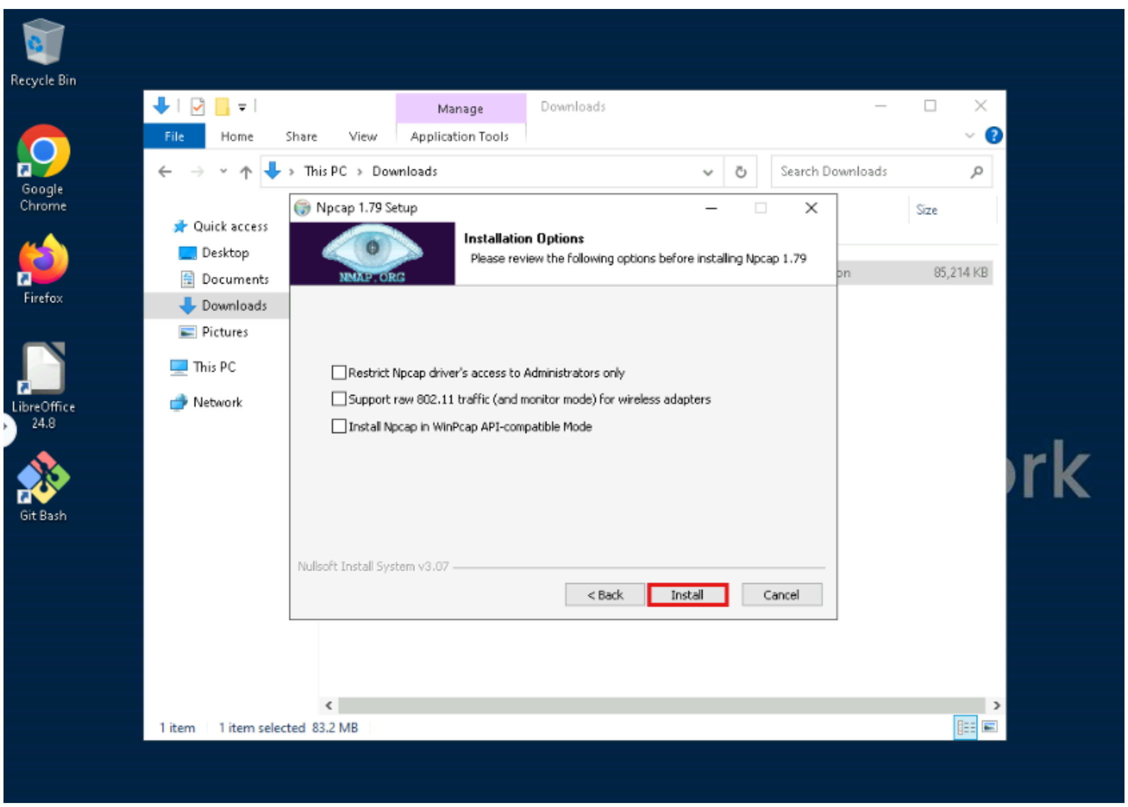
13. Click **Install**.



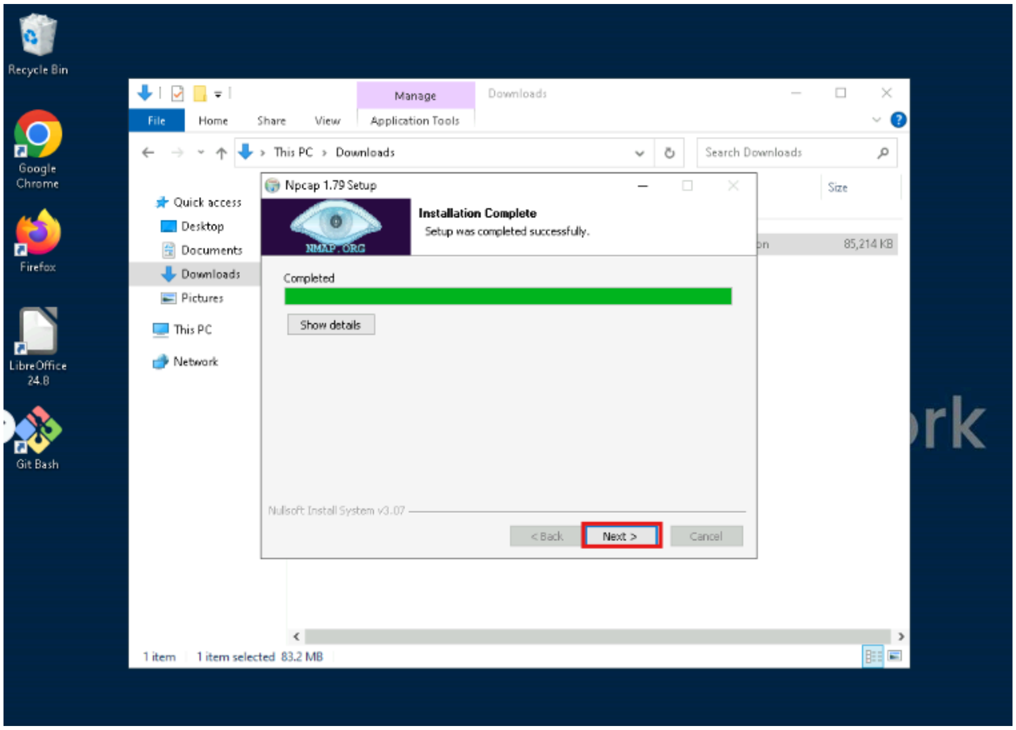
14. Installation may take several minutes.



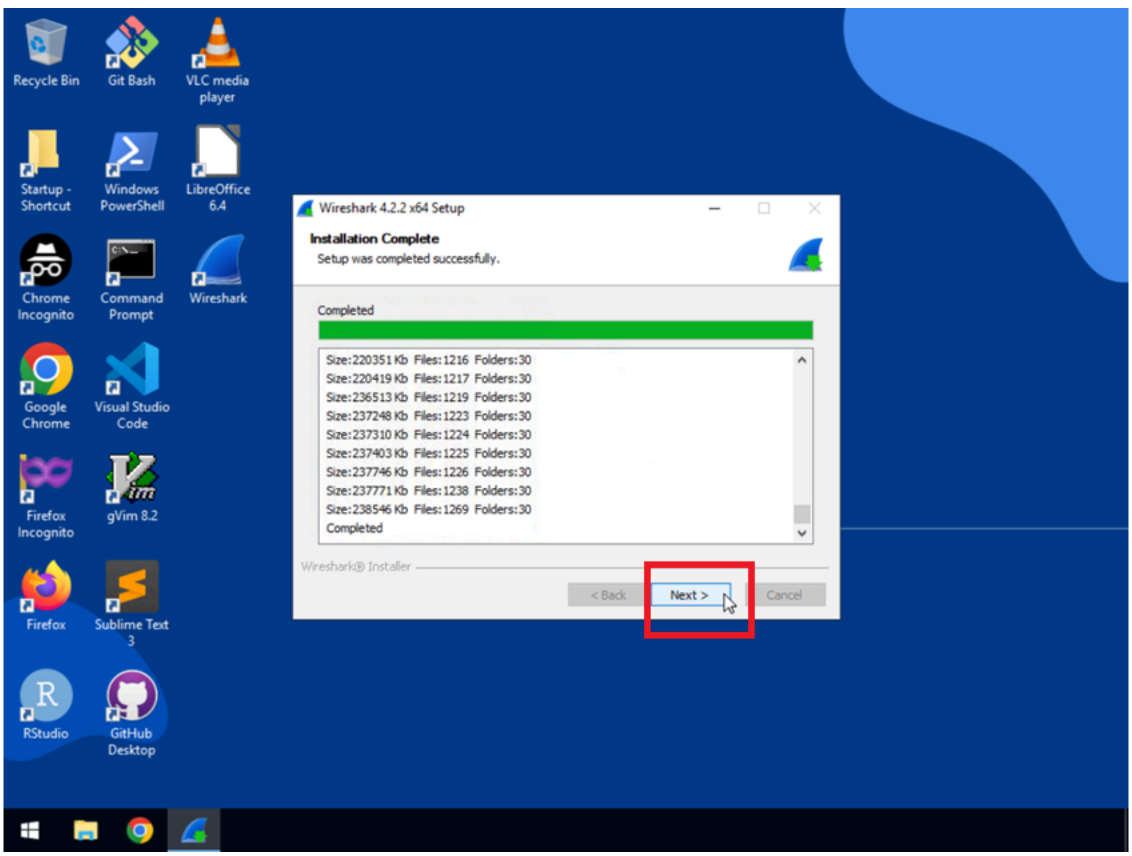
15. Click **Install**.



16. After the installation is complete, click **Next**, and then click **Finish**.

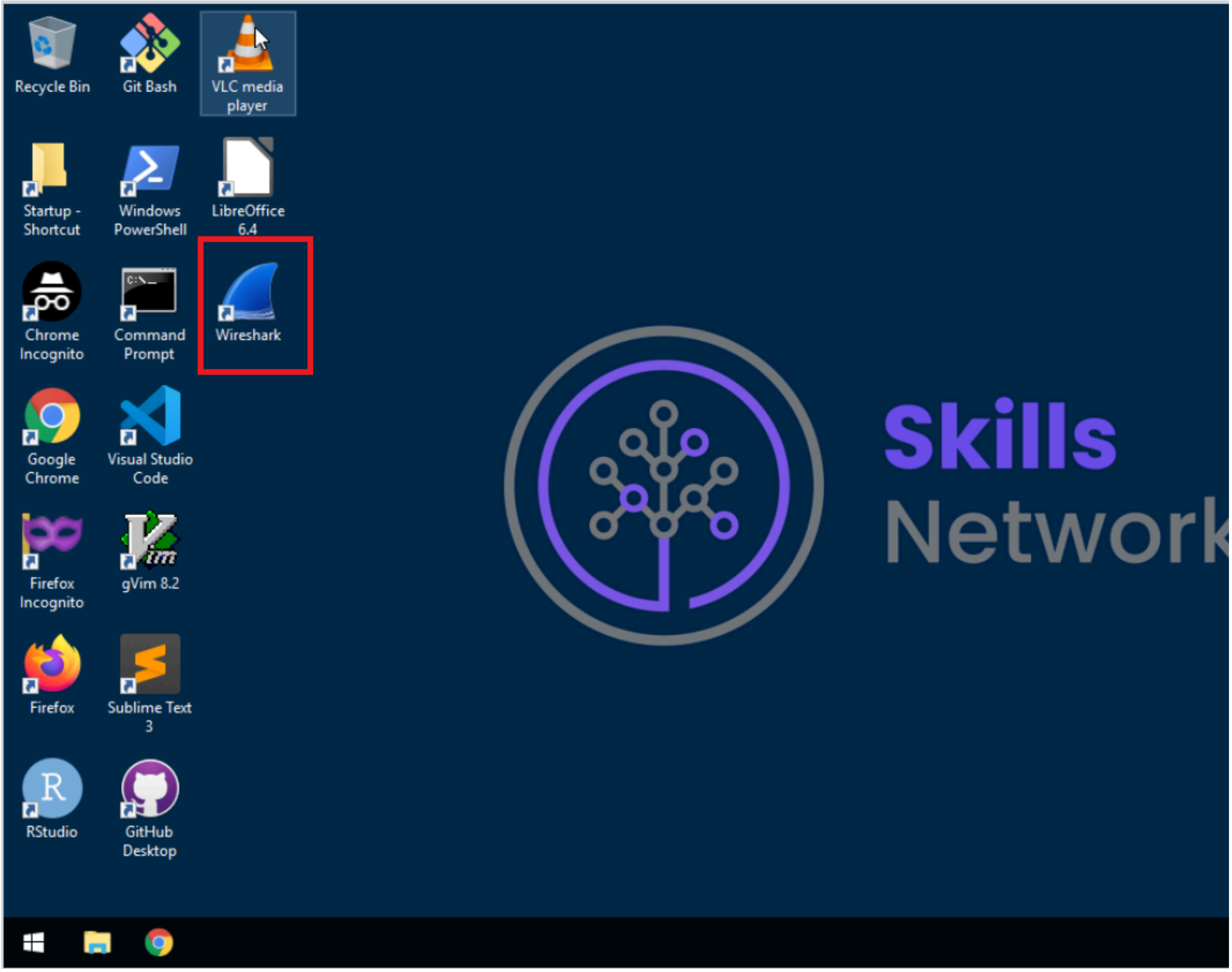


17. Once installation is complete, click **Next** to finalize installation.

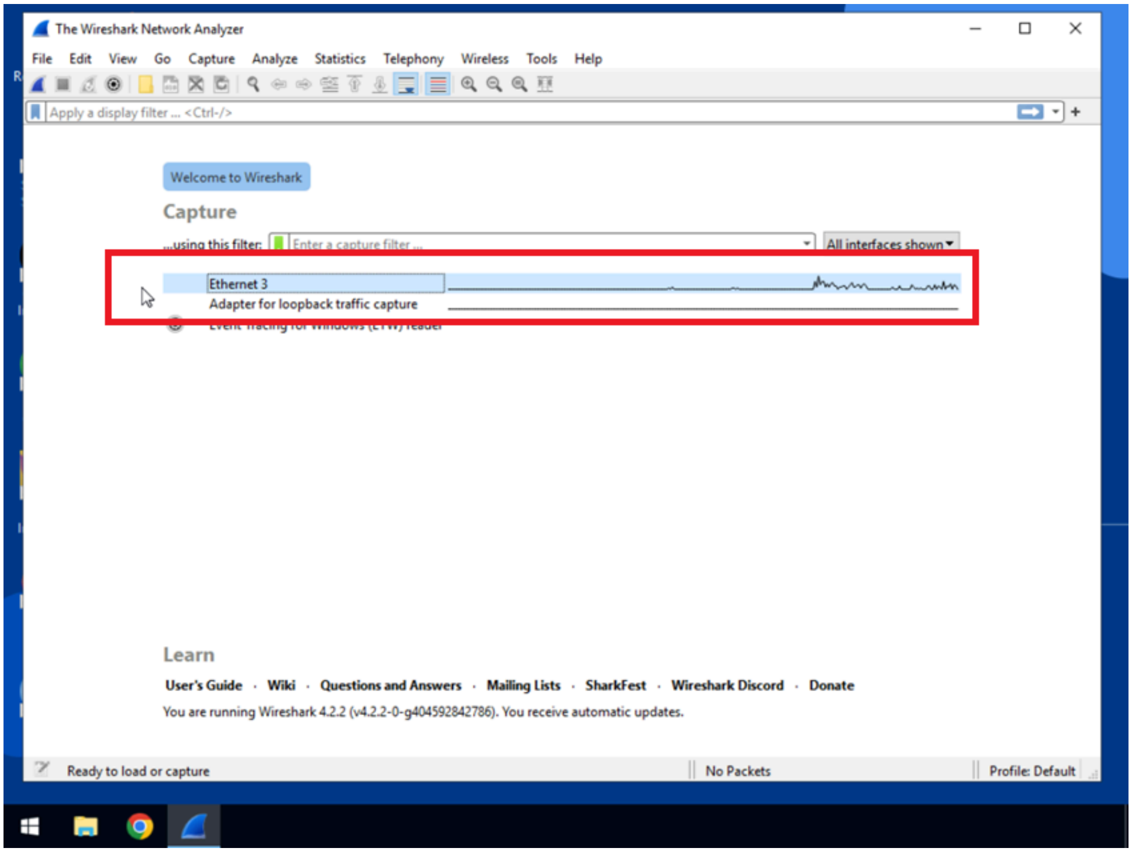


Exercise 2: Capture Packets using Wireshark

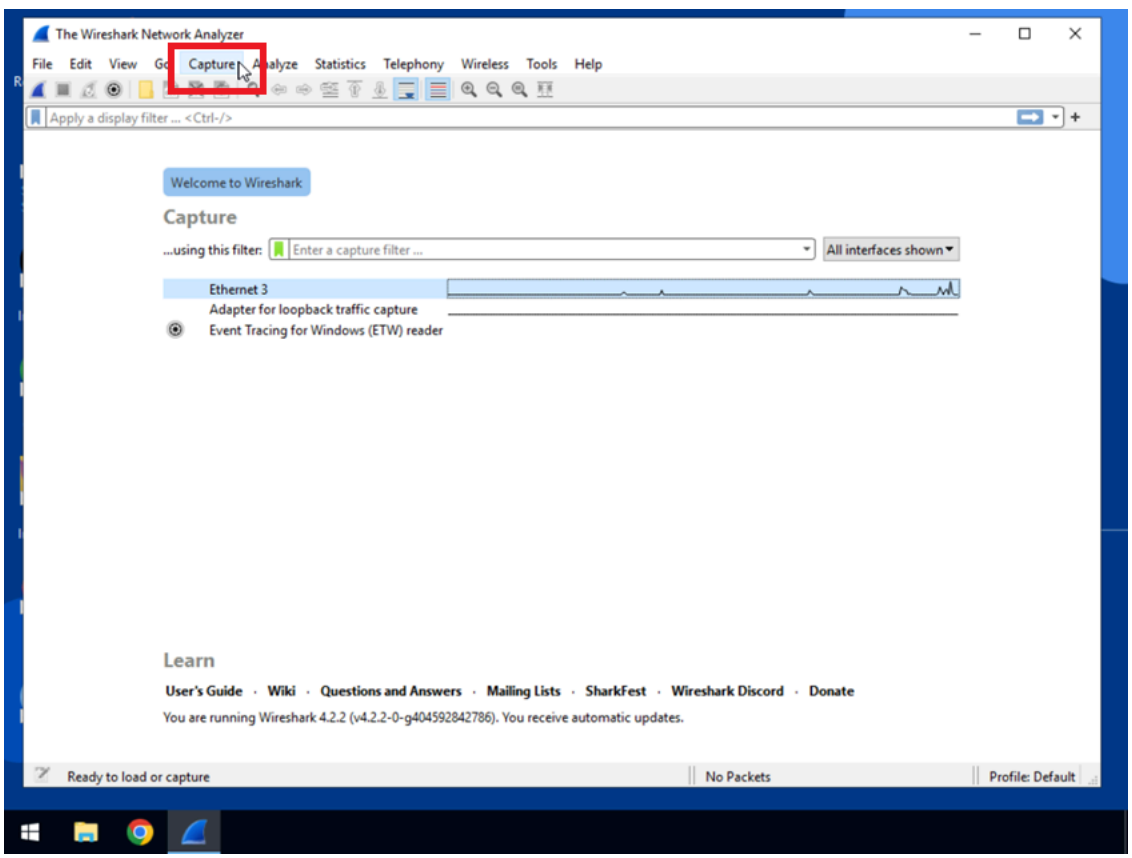
1. Double-click the **Wireshark** icon on the desktop to run the application.



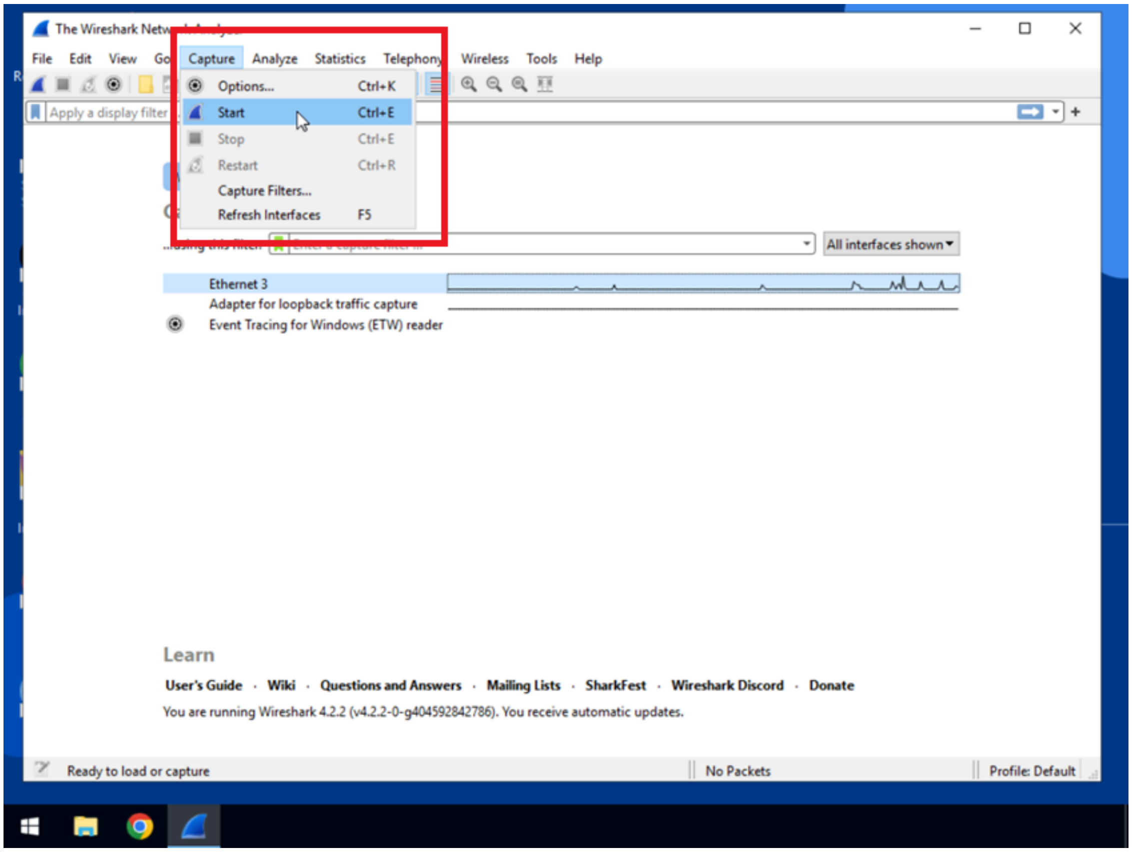
2. The first screen shows the network(s) available on this computer. Select and highlight **Ethernet 3**.



3. Click **Capture**.



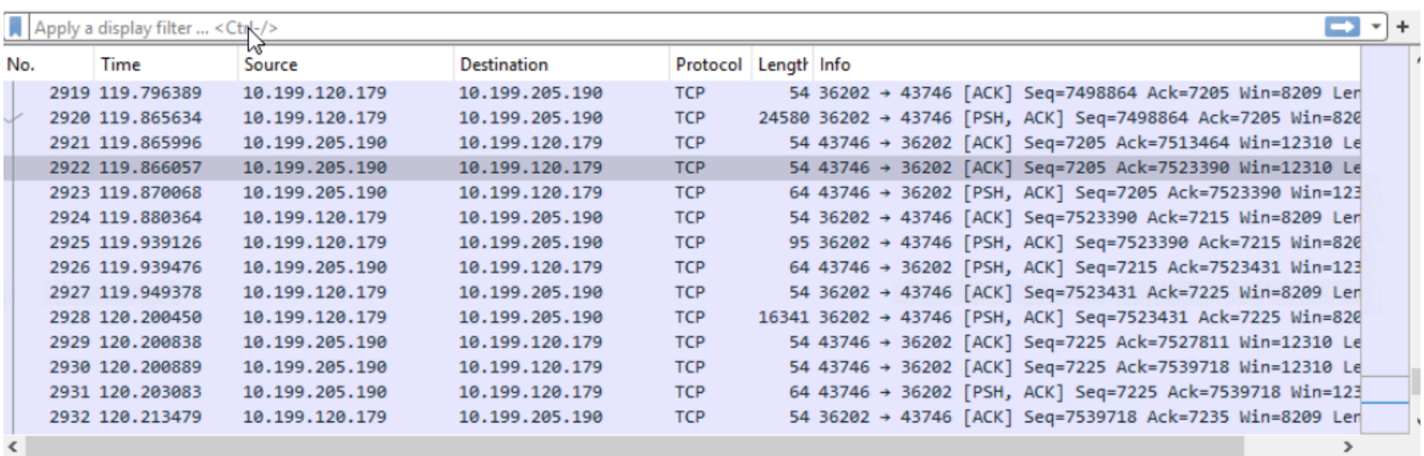
4. Select **Start** from the dropdown menu.



5. Wireshark will start to collect information about all packets that are sent and received. Click the Red **Stop Capturing Packets** icon located next to the shark fin.

Exercise 3: Review the Output

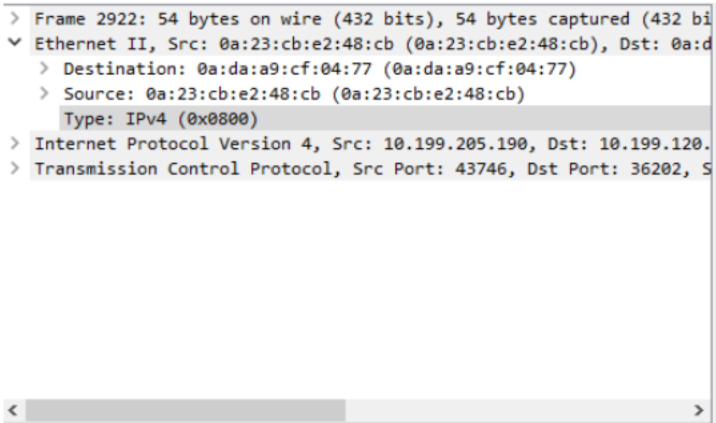
1. Familiarize yourself with the packet list pane.



No.	Time	Source	Destination	Protocol	Length	Info
2919	119.796389	10.199.120.179	10.199.205.190	TCP	54	36202 → 43746 [ACK] Seq=7498864 Ack=7205 Win=8209 Len
2920	119.865634	10.199.120.179	10.199.205.190	TCP	24580	36202 → 43746 [PSH, ACK] Seq=7498864 Ack=7205 Win=820
2921	119.865996	10.199.205.190	10.199.120.179	TCP	54	43746 → 36202 [ACK] Seq=7205 Ack=7513464 Win=12310 Le
2922	119.866057	10.199.205.190	10.199.120.179	TCP	54	43746 → 36202 [ACK] Seq=7205 Ack=7523390 Win=12310 Le
2923	119.870068	10.199.205.190	10.199.120.179	TCP	64	43746 → 36202 [PSH, ACK] Seq=7205 Ack=7523390 Win=123
2924	119.880364	10.199.120.179	10.199.205.190	TCP	54	36202 → 43746 [ACK] Seq=7523390 Ack=7215 Win=8209 Len
2925	119.939126	10.199.120.179	10.199.205.190	TCP	95	36202 → 43746 [PSH, ACK] Seq=7523390 Ack=7215 Win=820
2926	119.939476	10.199.205.190	10.199.120.179	TCP	64	43746 → 36202 [PSH, ACK] Seq=7215 Ack=7523431 Win=123
2927	119.949378	10.199.120.179	10.199.205.190	TCP	54	36202 → 43746 [ACK] Seq=7523431 Ack=7225 Win=8209 Len
2928	120.200450	10.199.120.179	10.199.205.190	TCP	16341	36202 → 43746 [PSH, ACK] Seq=7523431 Ack=7225 Win=820
2929	120.200838	10.199.205.190	10.199.120.179	TCP	54	43746 → 36202 [ACK] Seq=7225 Ack=7527811 Win=12310 Le
2930	120.200889	10.199.205.190	10.199.120.179	TCP	54	43746 → 36202 [ACK] Seq=7225 Ack=7539718 Win=12310 Le
2931	120.203083	10.199.205.190	10.199.120.179	TCP	64	43746 → 36202 [PSH, ACK] Seq=7225 Ack=7539718 Win=123
2932	120.213479	10.199.120.179	10.199.205.190	TCP	54	36202 → 43746 [ACK] Seq=7539718 Ack=7235 Win=8209 Len

Note the following: The packet list pane (the top pane) shows packets that were found. Every packet has its own row along with an assigned number. The following data points are included in this pane:
- **No.**: This is blank until you click on a packet. A checkmark will appear in the **No.** column to indicate other packets that are a part of the same conversation as the selected packet.
- **Time**: This indicates when the packet was captured. The default format for this entry is the number of seconds or partial seconds since the capture was created. This can be changed to a time-of-day format.
- **Source**: This indicates the IP address where the packet was sent from.
- **Destination**: This indicates the address that the packet was sent to.
- **Protocol**: This indicates the packet's protocol (i.e. TCP).
- **Length**: This indicates the packet length in bytes.
- **Info**: This indicates any additional details about the packet (if applicable).

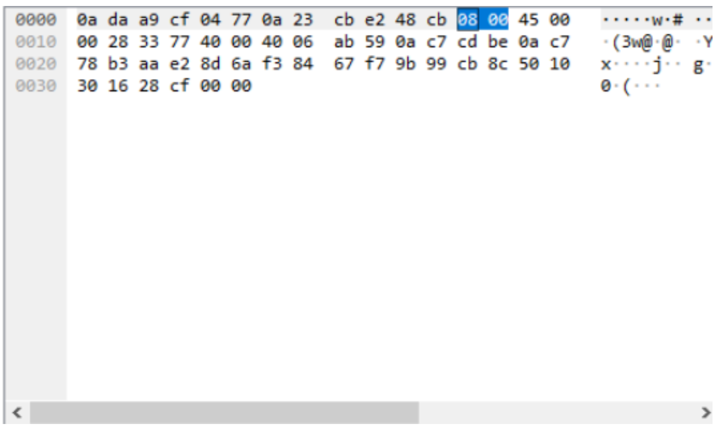
2. Familiarize yourself with the packet details pane (the bottom left pane).



This pane shows protocols and protocol fields of the selected packet. You can apply filters based on specific details by right-clicking on an item.

3. Familiarize yourself with the packet bytes pane (the bottom right pane).

   This pane shows the raw data of the selected packet. The data is in hexadecimal format (a number system using 0-9 and A-F). To change the format from hexadecimal to bits, you can right-click anywhere within the pane and select **as bits**.



Summary

In this lab you installed Wireshark, captured packets that were entering and exiting your network interface, and reviewed the resulting output.

Congratulations! You have completed this lab and are ready for the next topic.

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Skills Network